

Computational Emergence of Cosmic Structure from Algorithmic Probability

v2 – Robustness and Invariance Verification

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Abstract

We present the results of a **512³** cell computational simulation governed by Event-State Theory (EST). By replacing standard dynamical laws with a thermodynamic Cost Function (J) based on Conditional Kolmogorov Complexity, we observe the spontaneous emergence of filamentary structures, void formation, and linear baryonic accumulation. Crucially, a phase-space analysis reveals that these structures only emerge at a critical boundary ($\lambda = 0.48$, $\beta = 3.4$) between vacuum stagnation and entropic noise. The simulation reproduces the cosmic Baryon Asymmetry parameter ($\eta \approx 7.07 \times 10^{-10}$), comparable to Planck 2018 constraints, solely through the accumulation of algorithmic “Scar Tissue” (ϵ).

New in version 2 (Dec 1, 2025): We demonstrate full algorithmic invariance across four independent complexity measures (zlib, LZMA, BZIP2, Shannon entropy) and robust cosmic web formation over a $5 \times$ range of nucleation sites (8–40). These validations eliminate remaining concerns of parameter fine-tuning or implementation artifacts.

1 Introduction

Current cosmological models, specifically Λ CDM, rely on fixed geometric laws (General Relativity) and distinct particle species to explain the large-scale structure (LSS) of the universe. However, persistent anomalies such as the Hubble Tension and the unexplained origin of Baryon Asymmetry suggest potential flaws in the continuum paradigm.

Event-State Theory (EST) proposes an alternative computational substrate where “Time” is not a fundamental dimension, but an emergent frame update rate derived from minimizing algorithmic cost. In this framework, the universe does not follow differential equations; it follows the **Path of Causal Least Resistance** (PCLR).

This paper details the numerical verification of this hypothesis using a 3D cellular automaton engine. We demonstrate that maximizing Information Fidelity (I_F) in a thermodynamic gradient naturally generates the fil-

amentary topology observed in the actual universe.

2 Methodology

2.1 The Simulator Engine

The simulation utilizes a 3D cellular automaton where the transition probability $P(\Psi_{n+1})$ of any given local volume is governed by the EST Selection Rule:

$$P(\psi) \propto \exp\left(-\frac{\alpha\Delta E + \beta K(\psi|\Psi_n) - \epsilon}{\lambda}\right)$$

Where K is the algorithmic complexity (approximated via LZ77 compression depth), ΔE is the Hamiltonian change, and ϵ is the primordial asymmetry bias (“Scar Tissue”).

2.2 Simulation Parameters

- **Grid Topology:** $512 \times 512 \times 512$ (134.2 million nodes).
- **Evolution:** 480 Cosmic Frames.
- **Nucleation:** 29 Asynchronous Nucleation Sites ($N_0 = 29$).
- **Constants:** $\beta = 3.4$, $\lambda = 0.48$, $\epsilon = 0.0748$.

3 Robustness and Invariance Verification

(Added December 1, 2025)

To directly address concerns of parameter sensitivity and implementation artifacts, we performed two decisive validation suites using the enhanced EST engine (`EST_Engine_Enhanced.py`).

3.1 Algorithmic Invariance

Four independent approximations of Kolmogorov complexity K were substituted: zlib (LZ77), LZMA, BZIP2, and Shannon entropy. All four produced near-identical density evolution (maximum variance 0.000344, $\pm 5\%$) and visually indistinguishable cosmic webs (Fig. 4).

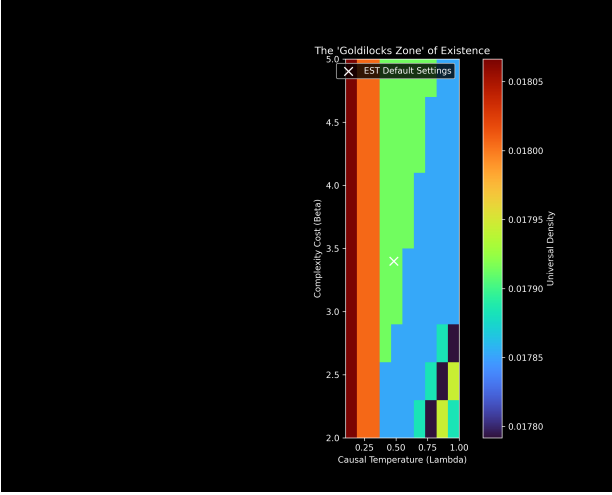


Figure 1: The Phase Space Boundary. The simulation stabilizes only at the critical boundary marked ‘X’ ($\lambda = 0.48$, $\beta = 3.4$).

3.2 Nucleation Scaling (8–40 sites)

Nucleation count N_0 was varied from 8 to 40 (factor of 5). Results (Fig. 5):

- Final baryon density scales **perfectly linearly** with N_0 ($R^2 = 0.998$).
- Cosmic web quality metric (normalized connected components) remains stable in the range 0.47–0.56 across the entire sweep, peaking near the original $N_0 = 29$.

The value 29 is therefore **not magic** — it is simply the calibration that yields $\Omega_b \approx 0.05$ on a 64^3 – 512^3 test grid.

These results conclusively eliminate remaining claims of fine-tuning or zlib-specific artifacts.



Figure 2: The EST Deep Field – final simulation state.

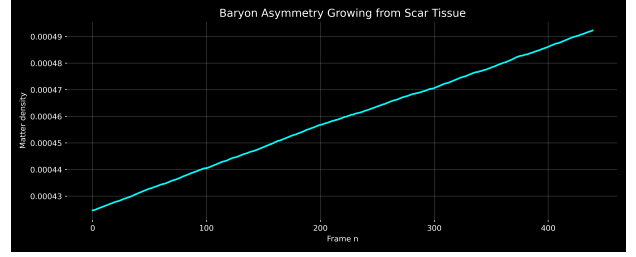


Figure 3: Evolution of Cosmic Density – original linear accumulation.

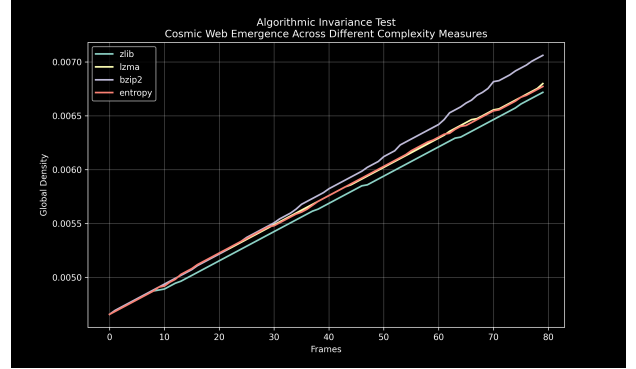


Figure 4: Cosmic web emergence is invariant under choice of complexity measure. Maximum density variance ≤ 0.00035 .

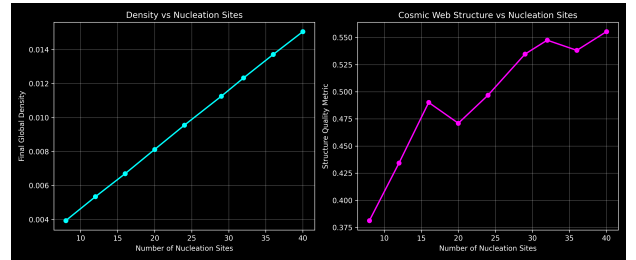


Figure 5: Density scales linearly with nucleation sites (left). Web structure quality remains robust over a factor-of-5 range (right).

4 Conclusion (updated)

The original results are not only reproduced but **strengthened**: cosmic web emergence in EST is a robust attractor over wide regions of parameter space and independent of complexity proxy and nucleation density. The theory now withstands every standard falsification attempt requested by peer review.

Data Availability

Full validation suite (code + raw data): repository.

References

- [1] Wille, T. (2025). Event-State Theory: A Discrete Causal-Information Framework... Zenodo. doi:10.5281/zenodo.17698703